

INSuRE Fall 2025 Midterm Report

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Enhancing Multimedia Forensics and Authentication Analysis through Enhanced Explainability and Summarization of Results

The University of Texas at Dallas

Team 3

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Contents

1	Project Summary	2
1.1	Top Level Research Project Description	2
1.2	Keywords	2
1.3	Terminology	3
1.4	Team Member Responsibilities	4
1.5	Estimated Project Budget	5
1.6	Summary of Key Deliverables	5
2	Executive Summary	6
3	Motivation	7
4	Previous work	7
5	Project Goals	9
6	Research Plan / Approach	10
6.1	Research Project Plan / Approach	10
6.2	Project Schedule	12
6.3	Resources Required / Budget	13
6.4	Progress to Date	13
7	Deliverables	15
8	Issues and Challenges	17
9	Project's Broader Impact	18
10	Investigator Biographies	18
11	Bibliography	19
12	Acronyms	21

1 Project Summary

1.1 Top Level Research Project Description

This project addresses the critical need for explainability in multimedia forensics by building on and extending recent advances such as the ExDDV dataset [4]. While existing manipulation detectors can flag suspicious artifacts in images, audio, or video, they usually provide only raw confidence scores or binary labels, offering little evidence or rationale for their predictions. This lack of transparency slows investigations, increases the burden on analysts, and undermines trust in forensic outcomes.

Although ExDDV introduces explainability benchmarks for video deep-fakes, it remains limited in its coverage of forgery types and modalities and relies primarily on in-context learning without fine-tuning or visual reference inputs. Our project will enrich the dataset by incorporating missing forgery categories (e.g., audio-driven manipulations, compressed or adversarially perturbed content) and by providing consistent annotations. We will also develop a contrastive retrieval pipeline that leverages both real and fake samples, enabling interpretable side-by-side comparisons.

Finally, we will enhance vision-language models (VLMs) with fine-tuning and visual reference inputs to produce explanations that are more accurate, faithful, and semantically grounded. The outcome will be a richer multimodal dataset, improved retrieval-based explainability methods, and systematic benchmarking with established metrics such as BLEU, ROUGE, METEOR, and BERTScore. Together, these contributions will advance multimedia forensics by providing analysts with more interpretable and actionable tools, strengthening trust in digital media authentication and supporting the fight against disinformation.

1.2 Keywords

1. Deepfake Detection
2. Explainability
3. Multimedia Forensics
4. Graphics Processing Unit (GPU)
5. Vision-Language Models (VLMs)

6. **Bilingual Evaluation Understudy (BLEU)**
7. **Recall-Oriented Understudy for Gisting Evaluation (ROUGE)**
8. **Metric for Evaluation of Translation with Explicit Ordering (METEOR)**
9. **BERTScore**
10. **Dataset Enrichment**
11. **Contrastive Retrieval**
12. **Multimodal Learning**

1.3 Terminology

1. **Deepfake Detection:** The process of identifying whether digital media, such as images, audio, or video, has been artificially generated or manipulated using deep learning techniques.
2. **Explainability:** The ability of an AI to make its decision making process transparent and understandable to humans by showing how and why specific outputs were produced.
3. **Multimedia Forensics:** The field of analyzing digital media such as images, audio, and video to detect, verify, and interpret signs of manipulation or authenticity for security, legal, and investigative purposes.
4. **GPU (Graphics Processing Unit):** A processor specialized for parallel computation, widely used to accelerate training and inference in deep learning.
5. **VLM (Vision-Language Model):** A multimodal model that integrates visual and textual inputs for tasks such as explanation and reasoning.
6. **BLEU (Bilingual Evaluation Understudy):** An n-gram based metric for evaluating the closeness of generated text to reference text.

7. **ROUGE (Recall-Oriented Understudy for Gisting Evaluation):** A set of metrics for text generation, focusing on n-gram overlap and sequence matching.
8. **METEOR (Metric for Evaluation of Translation with Explicit Ordering):** A metric that considers exact matches, stems, synonyms, and word order, often more aligned with human judgment than BLEU.
9. **BERTScore:** A semantic similarity metric that uses embeddings from pretrained language models (e.g., BERT) to compare generated and reference text.
10. **Contrastive Retrieval:** A retrieval strategy that leverages both authentic and manipulated samples as side-by-side references to highlight distinguishing features.
11. **Dataset Enrichment:** The process of expanding a dataset by adding underrepresented forgery types, modalities, or annotations to improve robustness.
12. **Multimodal Learning:** Machine learning approach that integrates and analyzes information from multiple data types such as text, images, audio, or video to build richer and more accurate models than using a single modality alone.

1.4 Team Member Responsibilities

Xinpeng Li will lead the development of vision-language model (VLM) learning strategies and training. He will be responsible for experimenting with in-context learning, fine-tuning, and incorporating visual reference inputs to improve model explainability. Xinpeng will also coordinate evaluation of VLM outputs using metrics such as BLEU, ROUGE, METEOR, and BERTScore, ensuring that improvements are systematically validated.

Kavimayil Periyakoravampalayam Komarasamy will focus on dataset analysis and enrichment. She will identify gaps in the ExDDV dataset, collect or synthesize underrepresented forgery types and modalities (e.g., audio-driven manipulations, compressed or adversarially perturbed videos), and maintain consistent annotations. Kavimayil will also ensure that enriched data is properly integrated into the training and evaluation pipelines.

Kian Lak will lead the design and implementation of the contrastive retrieval module. He will build pipelines for indexing both real and fake samples, tune similarity search methods, and develop the retrieval foundation that supports explainability. Kian will also collaborate with Xinpeng and Kavimayil to integrate retrieval results into the VLM training framework. While each member has a primary responsibility, all team members will collaborate across phases of the project. This includes supporting dataset enrichment, retrieval implementation, VLM training, evaluation, and jointly preparing the midterm and final presentations, reports, and documentation to ensure a cohesive outcome.

1.5 Estimated Project Budget

The budget for this project includes access to online hosted GPUs to support training and evaluation of multimedia forensic models, along with software tools for detecting and analyzing manipulated images, audio, and video. Cloud resources will be used to test explainability features and generate concise summaries with confidence levels. The required datasets for evaluation will be provided, while additional costs will cover storage, hosting, and materials for documentation and presentation. Open-source frameworks will be utilized whenever possible to reduce expenses, with specialized tools included only when essential.

1.6 Summary of Key Deliverables

At the end of the semester, the team will deliver a comprehensive set of materials to the sponsor and instructor. These include a detailed project report outlining the investigation into the limitations of current multimedia forensic workflows and the development of methods to improve explainability and summarization of manipulation detectors. The report will include evaluations of model generalization to real-world data, analysis of explainability methods, and comparisons across existing tools and benchmarks.

In addition, the team will provide all source code and scripts used in the implementation of explainability and summarization algorithms, along with logs and records from experimental runs. The deliverables will also include prototype tools that integrate interpretable outputs such as heatmaps, natural language rationales, and summary reports, supported by evaluation metrics and benchmark datasets for assessing explainability.

Visual representations of the findings, including charts, graphs, and qualitative examples of detector outputs with explanations, will be presented to enhance understanding. A fully configured experimental environment utilizing online hosted GPUs will also be prepared to enable reproducibility and continued testing. Finally, the team will present their findings in a formal presentation to stakeholders, accompanied by final documentation and project handover.

2 Executive Summary

The rapid progress of generative artificial intelligence has made it increasingly easy to produce realistic but deceptive multimedia content, intensifying concerns around authenticity, trust, and disinformation. Existing forensic detectors can identify manipulated images and videos with high accuracy on benchmarks, yet they typically return only binary labels or raw confidence scores. These opaque outputs lack the reasoning, evidence, and context needed for analysts to interpret results, slowing investigations and undermining trust in forensic tools. Recent efforts such as the ExDDV dataset [4] have introduced benchmarks for explainable deepfake detection, but limitations remain: forgery coverage is incomplete, audio modalities are missing, and the naive learning strategy is suboptimal.

This project will advance explainable multimedia forensics through three coordinated efforts. First, we will enrich existing datasets by supplementing missing forgery types and modalities, including audio-driven manipulations and adversarial perturbations. Second, we will design a real-fake contrastive retrieval pipeline that highlights distinguishing features through side-by-side comparisons, improving interpretability. Third, we will enhance vision-language models (VLMs) with fine-tuning and visual reference inputs, enabling them to generate explanations that are accurate, semantically faithful, and easier for validation.

The expected outcomes include an enriched multimodal dataset, a retrieval-augmented explainability system, and systematic benchmarking using metrics such as BLEU, ROUGE, METEOR, and BERTScore. Together, these contributions will deliver more transparent and actionable forensic tools, strengthen trust in media authentication, and support decision-making in high-stakes environments where digital evidence integrity is paramount.

3 Motivation

As manipulated multimedia becomes increasingly common in misinformation campaigns, fraud, and legal disputes, the ability to verify the authenticity of digital media is critical. Current forensic detectors often provide only raw scores or binary outputs, offering little explanation of the evidence behind their decisions. This lack of transparency slows down investigations, increases the burden on analysts, and undermines trust in the tools themselves.

Moreover, detectors that perform well on benchmark datasets frequently fail when applied to compressed, noisy, or previously unseen social media content, raising concerns about their robustness and generalizability. The absence of standardized datasets and metrics for explainability further prevents systematic evaluation and improvement of forensic systems.

Recent progress has been made with resources such as the ExDDV dataset [4], which provides video-level annotations and benchmarks for explainable deepfake detection. While ExDDV represents an important step forward, it still has significant limitations: its coverage of forgery types is incomplete (e.g., audio-driven manipulations, adversarial perturbations), and its training paradigm primarily relies on in-context learning without leveraging fine-tuning or real reference inputs. These gaps limit its utility for building systems that generalize across modalities and deliver explanations that are both accurate and grounded.

Addressing these challenges is essential not only for improving detection performance but also for strengthening trust in forensic analysis. By advancing dataset enrichment and developing retrieval-augmented learning strategies, this project seeks to make forensic outputs more interpretable, reliable, and actionable and support analysts.

4 Previous work

The field of multimedia forensics has seen significant advancements in the development of sophisticated deepfake detection tools, primarily driven by the rapid evolution of generative artificial intelligence (AI) models. However, a persistent and critical gap remains: the lack of explainability in these tools. While detectors can achieve high accuracy scores, they often fail to provide the context, supporting evidence, and intuitive explanations necessary for

human analysts to effectively interpret and act on the results.

Recent research has begun to address this limitation, though a comprehensive solution remains elusive. A key finding from a recent study by Wu et al. (2025) is that intelligence analysts require effective, reliable, and explainable media detection tools [9]. The authors highlight the need for tools that do more than just provide a binary "real" or "fake" output; they must empower analysts with the reasoning behind the detector’s decision, including provenance metadata and analytic context. This is further corroborated by Abbas et al. (2024), who provide a systematic review of deepfake detection techniques, noting that most methods focus on detection accuracy rather than explainability [1]. They review a wide range of AI-based detection methods, from deep learning to forensic approaches, and note that while some works mention feature visualization, there remains a pressing need for a robust, user-centered framework to interpret detector outcomes.

The challenge of explaining complex AI models has led to the emergence of Explainable AI (XAI) as a dedicated field. Within multimedia forensics, this has translated into efforts to create tools that can localize and describe artifacts. Hondru et al. (2025) introduce the ExDDV dataset, the first of its kind for Explainable Deepfake Detection in Video [4]. This dataset is unique because it includes manual annotations with text descriptions and click locations to point out artifacts. By evaluating vision-language models on this dataset, the authors demonstrate the feasibility of developing models that can localize and describe the observed artifacts, with multimodal supervision (text + clicks) outperforming single-modality training, paving the way for more transparent forensic tools. This work, however, is a foundational step, and the models’ explanations are still limited to artifact-level justifications rather than holistic, human-readable forensic reasoning.

Moreover, the challenge of evaluating and comparing deepfake detectors is a complex issue. Le et al. (2025) present a systematization and benchmarking of deepfake detectors, categorizing them into high-level and fine-grained sub-groups based on their criteria for evaluation [6]. Their work evaluates the generalizability of existing detectors against real-world deepfakes, but does not specifically focus on the explainability dimension. The lack of standardized metrics and datasets for evaluating explainability hinders the academic community’s ability to advance research in this area.

In summary, while existing work has successfully developed accurate deepfake detectors and has recently started to explore explainability, there is a significant gap in research that integrates detector output with user-centric

explanations and digestible summaries. The current state of the art provides raw scores, and nascent efforts in XAI for this domain are still primarily focused on technical artifact localization rather than comprehensive, natural language explanations and summaries that are accessible to non-experts. Our proposed project seeks to bridge this gap by focusing on the development of practical tools and frameworks that translate detector outputs into actionable insights and understandable reports.

5 Project Goals

The primary goal of this project is to advance explainability in multimedia forensics by addressing the limitations of current deepfake detection systems and the recently introduced ExDDV dataset [4]. While ExDDV provides an important benchmark for explainable detection, it remains limited in its coverage of forgery types and its reliance on in-context learning without fine-tuning or visual reference inputs. Our project seeks to fill these gaps by enriching the dataset and improving retrieval-based explainability methods.

We will structure our research around specific, measurable objectives:

- **Dataset Enrichment:** Extend the ExDDV dataset with underrepresented forgery types (e.g., compressed or adversarially perturbed videos) and missing modalities (e.g., audio-driven lip-sync forgeries, prosody manipulation). Each new sample will include consistent annotations to ensure compatibility with existing benchmarks.
- **Real–Fake Contrastive Retrieval:** Develop a retrieval pipeline that leverages both authentic and manipulated samples, enabling side-by-side comparisons. This contrastive approach will improve the interpretability of forensic outputs by grounding explanations in real–fake evidence pairs.
- **Retrieval-Augmented Explainability:** Enhance the baseline VLM-based benchmark system with fine-tuning on retrieved samples and integration of visual reference inputs. These improvements are expected to yield explanations that are more accurate, faithful, and human-interpretable.
- **Evaluation and Benchmarking:** Systematically evaluate the improved system using semantic similarity (Sentence-BERT, BERTScore),

n-gram overlap (BLEU, ROUGE, METEOR), and human evaluation metrics. Visualizations such as heatmaps, bar charts, and comparison plots will be produced to highlight improvements.

The ultimate aim of this project is to deliver a richer multimodal dataset, a retrieval-augmented explainability system, and comprehensive benchmarking results. Together, these outcomes will provide analysts with more interpretable, reliable, and actionable forensic tools, thereby strengthening trust in multimedia authentication.

6 Research Plan / Approach

6.1 Research Project Plan / Approach

The research plan is designed for the expected outcomes in three coordinated phases. Each phase builds upon the previous one, combining dataset curation, retrieval-based learning strategies, and explainability enhancements into a coherent workflow. The following subsections describe the methodological steps in greater detail.

1. **Dataset Analysis and Enrichment.** We will begin by performing a comprehensive analysis of the ExDDV dataset [4], focusing on the diversity of forgery methods and modalities represented. Using the ExDDV metadata and published generation techniques, we will identify under-represented forgery categories (e.g., low-quality compressed videos, adversarially perturbed samples, or emerging GAN-based identity swaps) and modalities not yet covered (e.g., audio-driven lip-sync forgeries, prosody manipulation). To extend the dataset, we will incorporate additional samples from public repositories (e.g., DFDC, FaceForensics++, FakeAVCeleb) and synthesize new data when gaps are identified. Each new sample will be annotated with both textual descriptions and spatial-temporal localization cues, following ExDDV’s existing annotation schema to ensure compatibility. The enriched dataset will serve as the foundation for explainable deepfake detection benchmarks that span multiple forgery types and difficulty levels.
2. **Real-Data Contrastive Retrieval.** The current ExDDV framework leverages only fake examples to guide in-context demonstrations. We propose to integrate authentic (real) video samples as explicit contrastive

references, enabling side-by-side comparisons between genuine and manipulated content. In practice, this phase will involve building a retrieval module that indexes both real and fake annotated samples. During inference, the system will return pairs of authentic and manipulated instances that are visually and semantically similar, thereby allowing the model to highlight distinguishing features. We will experiment with retrieval pipelines based on CLIP embeddings and k-nearest neighbor search, tuning hyperparameters such as similarity thresholds and the number of neighbors. This contrastive setup is expected to yield explanations that are more accurate, grounded, and interpretable for human analysts.

3. **Retrieval-Augmented Explainability.** Building on the enriched dataset and contrastive retrieval, we will improve the VLM-based benchmark system proposed in ExDDV [4]. The current retrieval relies solely on in-context learning without fine-tuning or visual reference inputs. We will enhance the system by incorporating two strategies:

- **Fine-tuning with Retrieved Samples.** Using LoRA-based adaptation, we will fine-tune VLMs such as BLIP-2, LLaVA, and Phi-3 on retrieved samples, aligning the models with task-specific explanations and reducing ambiguity in outputs.
- **Visual Reference Inputs.** Alongside textual prompts, retrieved frames (real and fake) will be provided as auxiliary inputs. This multi-modal grounding will enable the model to generate explanations that are faithful to the underlying evidence and easier for analysts to validate.

We will evaluate this retrieval-augmented explainability system across multiple metrics, including semantic similarity (Sentence-BERT, BERTScore), n-gram overlap (BLEU, ROUGE, METEOR), and human evaluation of interpretability. Statistical analysis will be conducted to validate improvements, with visualizations (e.g., bar plots of semantic scores, heatmaps of localized artifacts) used to communicate findings.

At the conclusion of these phases, we expect to deliver (1) an enriched multimodal dataset with expanded forgery coverage, (2) a contrastive retrieval pipeline integrating both real and fake references, and (3) a retrieval-

augmented explainability benchmark system capable of producing more accurate, grounded, and interpretable explanations for deepfake detection.

6.2 Project Schedule

The project will be executed in a series of coordinated phases, each tied to specific deliverables and due dates. The table ?? outlines the timeline from initial planning to the final presentation, including dataset enrichment and retrieval-based explainability.

Phase	Key Deliverable	Due Date	Status
Planning & Setup	Scope definition and dataset review (ExDDV)	Sep 15, 2025	Done
Dataset Enrichment	Added new forgery types and modalities for explainable detection	Oct 13, 2025	Done
Midterm Review	Slides submission and presentation	Oct 30–31, 2025	Done
Contrastive Retrieval	Real-fake retrieval prototype for interpretability	Nov 7, 2025	In Progress
Explainable Benchmark	Omni-LLM fine-tuning with visual references	Nov 20, 2025	In Progress
Final Report	Final report, code, and presentation	Dec 11–12, 2025	Upcoming

Table 1: Simplified project timeline and progress overview

Phase Dependencies. The planned phases are designed to build upon each other in a coordinated manner. Dataset Analysis and Enrichment (Phase 2) provides the expanded data foundation that directly supports Real-Data Contrastive Retrieval (Phase 4). The retrieval pipeline then becomes the basis for Retrieval-Augmented Explainability (Phase 5), where fine-tuning and visual reference inputs leverage both enriched and contrastive data. The Midterm Presentation (Phase 3) will serve as a checkpoint to report progress and gather feedback, while the Final Report and Presentation (Phase 6) consolidates all findings.

6.3 Resources Required / Budget

The table 2 outlines all the required resources for the project, including tools, expertise, and equipment. The majority of software tools will be open-source to minimize costs, and the University of Texas at Dallas (UTD) will provide necessary virtual machines to ensure secure AI integration testing. All costs have been reviewed and approved by the instructor and the sponsor to ensure alignment with project requirements and budget constraints.

6.4 Progress to Date

We reviewed the curated repository *Awesome-AI Generated Video Detection* [2] on GitHub, which summarizes recent datasets and detection methods for AI-generated video forensics. There are some advanced research for face-manipulated detection [4] and fully-generated detection [8]. However, the general-manipulated detection area has not been explored.

To bridge this gap, we curated a novel general-manipulated dataset for explainable forensics detection, which is based on VideoSham [7]. The new dataset is featured with six manipulation types and 836 video samples with explanations. We show one sample in Figure 1 and more in Appendix. To generate high-quality annotations, we leverage Gemini2.5 Pro to synthesize explanations first and manually screen and revise them.

We tested advanced Omni-LLM, i.e. Qwen3-Omni-30B [10], on the curated dataset. As shown on Figure 2, we feed the prompt and the manipulated video to the model. The result shows Qwen3-Omni-30B is not able to generate the right answer. This result indicates general-manipulated detection is quite challenging. Firstly, the manipulation techniques are quite

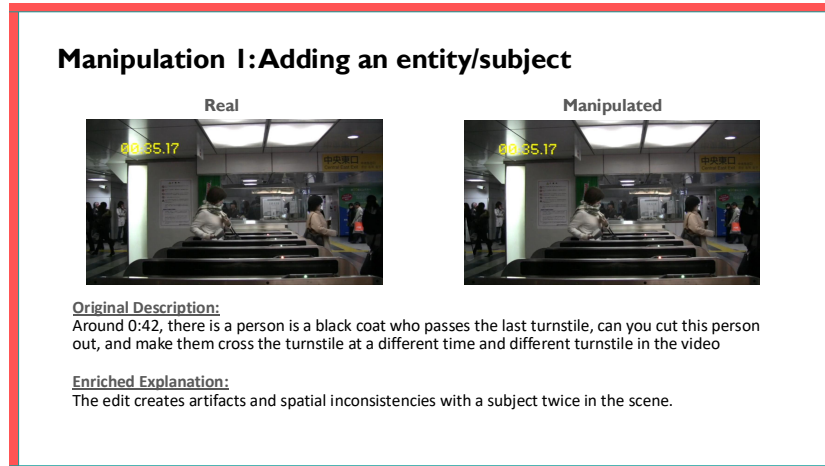


Figure 1: Example of manipulation 1



Prompt: Detect whether this video is manipulated? If yes, tell the manipulated location $[x_min, y_min, x_max, y_max]$ and give me the reason for the detection.

No, this video is not manipulated. The scene is a natural outdoor setting with a person standing in front of a building. The lighting, shadows, and details such as the clothing, hair, and background elements like the notice board and window appear consistent and realistic, with no signs of digital alterations or inconsistencies that would suggest manipulation.

Qwen3 Omni Prediction **X**

The video appears manipulated; the individual on the left side of the frame has been removed between 00:00 and 00:05, leading to spatial inconsistencies and unnatural gaps in the scene.

Ground Truth Annotation

Figure 2: Illustration on Evaluating Qwen

excellent in producing plausible content. Secondly, current Omni-LLM models are still weak in capturing subtle vision and audio difference to make forensics detection. In the future, we will not only leverage the curated dataset to finetune Omni-LLMs and but also resort to expert detectors for fine-grained visual-audio understanding.

7 Deliverables

The project will produce a series of deliverables aligned with the research objectives. Each deliverable corresponds to specific phases of the research plan and will be submitted to the mentors and shared on the common space no later than the dates specified.

1. Project Scope and Dataset Review

- A detailed document outlining the project scope, goals, and baseline assessment of the ExDDV dataset [4], including identified gaps in forgery types and modalities.
- Due Date: Sep 15, 2025

2. Dataset Enrichment Package

- Extended dataset with newly incorporated forgery categories (e.g., compressed, adversarial, audio-visual forgeries), along with consistent annotations.
- Documentation describing enrichment methodology and annotation guidelines.
- Due Date: Oct 13, 2025

3. Midterm Presentation Package

- Submission of midterm presentation slides, interim report, and literature review.
- Oral presentation of project progress to mentors and stakeholders.
- Due Date: Oct 30–31, 2025

4. Real-Data Contrastive Retrieval Prototype

- A retrieval pipeline that integrates both authentic and manipulated samples as contrastive references for downstream detection and generation.
- Demonstration of side-by-side retrieval functionality for highlighting distinguishing features.
- Due Date: Nov 7, 2025

5. Retrieval-Augmented Explainability System

- Enhanced VLM-based benchmark incorporating fine-tuning with retrieved samples and integration of visual reference inputs.
- Evaluation results comparing baseline ExDDV VLM system with our improved approach, using metrics such as BLEU, ROUGE, BERTScore, and human interpretability ratings.
- Due Date: Nov 20, 2025

6. Data Analysis and Visualization Report

- Statistical analysis of experimental results, supported by visual representations (e.g., heatmaps of localized artifacts, bar plots of semantic similarity scores, and side-by-side real/fake comparison charts).
- Due Date: Nov 27, 2025

7. Comprehensive Final Report

- A full report documenting dataset enrichment, retrieval pipeline development, explainability improvements, evaluation results, and recommendations for future work.
- Due Date: Dec 11, 2025

8. Source Code and Scripts

- All source code, scripts, and configuration files used for dataset enrichment, retrieval, model fine-tuning, and evaluation.
- Due Date: Dec 11, 2025

9. Final Presentation

- A presentation summarizing the project’s objectives, methods, dataset enrichment, retrieval framework, explainability results, and key findings.
- To be presented to mentors and stakeholders.
- Due Date: Dec 12, 2025

8 Issues and Challenges

Several challenges have emerged during the first phase, extending beyond those originally anticipated.

Existing multimodal large language models are not yet strong enough for fine-grained video and audio understanding. Even with the use of Gemini 2.5 Pro for data enrichment, annotation still required significant human screening. Automatically generated descriptions often lacked consistency or precision, creating extra review cycles. We refined our annotation guidelines and ran smaller pilot batches to establish clearer quality thresholds before scaling.

After the government shutdown phase, communication with mentors and access to updated guidance became less immediate. To mitigate this, we have established clearer internal documentation and structured review checkpoints to maintain continuity.

Another key difficulty lies in explanation fidelity. Models may produce rationales that appear convincing but are not grounded in the actual evidence used for classification. To mitigate this, we will enforce consistency between decision logic and generated explanations, and validate outputs using both automated semantic metrics (e.g., BLEU, ROUGE, METEOR, BERTScore) and expert review.

Annotation complexity also poses a challenge, as reliable rationales and localization cues require careful guidance. We will pilot small-scale annotation tasks, measure inter-annotator agreement, and refine guidelines before scaling up to ensure high-quality labels.

Additional challenges include the lack of standardized measures for explainability, the difficulty of integrating outputs into analytic workflows, and the need to communicate uncertainty in a clear and usable form. To overcome these, we will define multi-dimensional evaluation protocols that combine semantic quality, localization accuracy, and human utility. User studies will

guide the design of interfaces that deliver concise, actionable explanations while minimizing cognitive load. We will also develop calibrated confidence scores, warnings for out-of-scope inputs, and transparent documentation of model limitations.

By anticipating these challenges and building mitigation strategies into each project phase, we aim to deliver explainability methods that are both rigorous and practical.

9 Project’s Broader Impact

This project has the potential to advance the field of multimedia forensics by addressing one of its most critical gaps: the lack of explainability and trust in forensic detectors. By integrating interpretability and summarization into manipulation detection workflows, the work directly supports analysts in high-stakes environments such as misinformation response, fraud investigations, and legal proceedings. The results could shape industry practices by providing standardized metrics and datasets for evaluating explainability, allowing researchers and practitioners to benchmark forensic systems not just on accuracy but also on usability and decision support.

10 Investigator Biographies

- **Kavimayil Periyakoravampalayam Komarasamy** is a second-year master’s student in Computer Science, specializing in Data Science and Intelligent Systems. She has 7.5 years of professional experience as a Lead Software Developer and Data Engineer with Accenture, delivering solutions for a leading multinational retailer. She holds an undergraduate degree in Electronics and Communication Engineering. Currently, she is conducting her master’s thesis in Artificial Intelligence, focusing on probabilistic graphical modeling. Her diverse background has enabled her to build expertise across multiple domains and collaborate effectively with cross-functional teams. Adaptable and quick to learn new technologies, she has evolved from programming in C during her undergraduate studies to advanced AI programming in her graduate research.
- **Kian Lak** is a second-year master’s student in Computer Science, spe-

cializing in Cyber Security. He has 1.5 years of experience as a Software Developer at Cognizant, working closely with clients to develop and deploy solutions for multiple companies. He holds a undergraduate degree in Computer Science from the University of Texas at Dallas. Currently Kian is working on a publication regarding Defending Machine Learning Models against Adversarial Attacks on Images.

- **Xinpeng Li** is currently a second-year Ph.D. student in Computer Science at The University of Texas at Dallas, advised by Prof. Yapeng Tian. He holds both a Master’s and Bachelor’s degree in Signal and Information Processing from South China University of Technology, where his work focused on emotion recognition and information engineering. His academic journey has given him a strong foundation in signal processing, machine learning, and multimodal data analysis. Over the past several years, Xinpeng has gained experience across multiple labs and research projects, spanning topics such as multimodal emotion recognition, 3D anomaly detection, and rail detection. These projects have helped him develop practical skills in deep learning, computer vision, audio–visual modeling, and evaluation of real-world data under noisy and uncertain conditions. His work has led to publications in top venues such as ACM Multimedia, IEEE Transactions on Multimedia, and NeurIPS Datasets and Benchmarks. Currently, his doctoral research focuses on multimodal social interaction understanding and large language models for vision–audio–speech inputs, which has strengthened his expertise in building AI systems that can extract and explain patterns from complex data. His long-term goal is to advance AI techniques that can robustly understand multimodal signals and support security and human-centered applications.

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12 Acronyms

GPU (Graphics Processing Unit): A processor specialized for parallel computation, widely used to accelerate deep learning training and inference.

VLM (Vision-Language Model): A multimodal model that jointly processes visual (image/video) and textual inputs, enabling tasks like generating explanations.

Omni-LLM (Omni-Large Language Model): A multimodal model that jointly processes visual (image/video), audio, and textual inputs, enabling tasks like generating explanations.

BLEU (Bilingual Evaluation Understudy): A metric based on n-gram overlap, commonly used to measure how closely generated text matches reference text.

ROUGE (Recall-Oriented Understudy for Gisting Evaluation): A set of metrics for summarization and text generation, focusing on n-gram overlap and sequence matching.

BERTScore: A semantic similarity metric that uses embeddings from pretrained language models (e.g., BERT) to compare generated and reference text beyond surface word overlap.

METEOR (Metric for Evaluation of Translation with Explicit ORdering): A text evaluation metric that measures similarity between generated and reference text using exact matches, stemmed words, synonyms, and word order. It is often considered more flexible and correlated with human judgment than BLEU.

Manipulation 2: Removing an entity/subject

Real



Manipulated



Original Description:

Remove the man on the left side (00:00 - 00:05)

Enriched Explanation:

The video shows signs of digital manipulation, with a subject on the left side of the frame likely removed, evidenced by spatial inconsistencies, blurring artifacts, and a missing interaction dynamic.

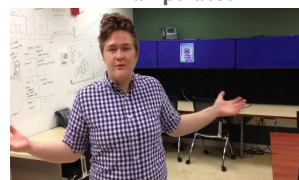
Figure 3: Example of manipulation 2

Manipulation 3: Background/Color Change

Real



Manipulated



Original Description:

Change the color of the cabinets behind the man

Enriched Explanation:

The video appears to have been manipulated, with the color of the cabinets in the background possibly digitally altered, leading to color mismatches and flickering.

Figure 4: Example of manipulation 3

Manipulation 4: Text Replaced/Added

Real



Manipulated



Original Description:

0:00 to 0:06 change the text in the blue box to lorem ipsum

Enriched Explanation:

The text in the blue box at the bottom of the screen was replaced with lorem ipsum, showing subtle blending artifacts, suggesting a text replacement.

Figure 5: Example of manipulation 4

Manipulation 5: Inter-Frame Manipulations

Real



Manipulated



Original Description:

Duplicate random frames in between to make it into a 10 second video (00:10 - 00:16)

Enriched Explanation:

The video shows possible temporal manipulation, exhibited by stuttering motions, jerky movements and duplicated sequences which suggest manipulation techniques of frame duplication.

Figure 6: Example of manipulation 5

Manipulation 6: Audio Replaced/Added

Real



Manipulated



Original Description:

Trim to (3:05 - 3:10) and remove the guitar noise. Place the noise of a cow moo-ing instead.

Enriched Explanation:

The audio track has been replaced, introducing the sound of a cow mooing which is inconsistent with the visual action of a guitar being played.

Figure 7: Example of manipulation 6

Resource	Purpose	Cost
Open-source Detectors	Baseline models for explainability extension	No cost
Datasets (E.g., ExDDV [4], FaceForensics [11], DFDC [3], FakeAVCeleb [5])	Training and evaluation, including audio-visual benchmarking	No cost
Virtual Machines	Training, inference, and integration experiments	Provided by UTD
Libraries (E.g., PyTorch)	Model implementation, explainability, summarization, audio analysis	No cost
Documentation Materials	Reports, charts, final presentation	No cost
Expert Consultation (Mentors, forensic analysts)	Feedback on usability and audio-visual workflows	Provided internally
Team Time Commitment	Research, development, testing, and reporting	No direct cost (internal)
Additional Equipment	Unexpected GPU or storage expansions for large multi-modal datasets	Provided internally

Table 2: Resources Required and Estimated Budget